# Yohan Abeysinghe

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## EDUCATION

**University of Maryland, College Park, USA** **Aug 2025 – Present**
*Ph.D. in Electrical and Computer Engineering* *GPA: 3.85* *Advisor: Prof. Dinesh Manocha*

**University of Moratuwa, Sri Lanka** **Jan 2020 – Jul 2024**
*B.Sc. (Hons) in Electronics and Telecommunication Engineering* *GPA: 3.89* *First Class Honours*

**Maliyadeva College, Kurunegala, Sri Lanka** **Dec 2018**
*GCE Advanced Level: Mathematics, Physics, Chemistry* *3 A's* *Z-score: 2.78* *Island Rank: 24*

## PUBLICATIONS

- Gershom Seneviratne, **Yohan Abeysinghe**, Jianyu An, Vaibhav Shende, Rahul Kumar, Dinesh Manocha. "VEGA: Learning Navigation VLAs from In-the-Wild Egocentric Video with Geometric Trajectory Supervision." *arXiv preprint arXiv:2606.18426*, 2026. [ArXiv] [Project Page]
- **Yohan Abeysinghe**, Muhammad Akhtar Munir, Sanoojan Baliah, Ron Sarafian, Fahad Shahbaz Khan, Yinon Rudich, Salman Khan. "Synergistic neural forecasting of air pollution with stochastic sampling." *npj Clean Air*, vol. 2, article 36, 2026. [Paper] [arXiv]
- Sanjoy Chowdhury, Mohamed Elmoghany, **Yohan Abeysinghe**, Junjie Fei, Sayan Nag, Salman Khan, Mohamed Elhoseiny, Dinesh Manocha. "MAGNET: A Multi-agent Framework for Finding Audio-Visual Needles by Reasoning over Multi-Video Haystacks." In *Advances in Neural Information Processing Systems (NeurIPS)* 2025. [Project Page]
- Sanoojan Baliah, **Yohan Abeysinghe**, Rusiru Thushara, Khan Muhammad, Abhinav Dhall, Karthik Nandakumar, Muhammad Haris Khan. "VFace: A Training-Free Approach for Diffusion-Based Video Face Swapping." In *Winter Conference on Applications of Computer Vision (WACV) 2026*.

## RESEARCH EXPERIENCE

**University of Maryland, College Park, USA** **Aug 2025 – Present**
*Doctoral Researcher* *Advisor: Prof. Dinesh Manocha*

- Trained and deployed VEGA vision-language-action models for robot navigation, translating policies learned from geometrically supervised egocentric video into real-robot execution.
- Currently developing a self-supervised variant of VEGA for learning robot navigation policies.
- Developing runtime steering for navigation policies using Euclidean signed distance fields and dynamic-obstacle velocity estimates; evaluating goal reaching and obstacle avoidance through real-robot experiments and controlled ablations.

**Mohamed bin Zayed University of Artificial Intelligence, UAE** **July 2024 – July 2025**
*Research Associate* *Advisor: Dr. Salman Khan*

- Developed SynCast, a spatiotemporal air-pollution forecasting framework combining a pretrained weather transformer, LoRA-based regional adaptation, and diffusion-based stochastic refinement.
- The model reduced 24-hour $PM_{2.5}$ forecasting RMSE by approximately 18–39% against Aurora and AirCast across evaluated resolutions in the Middle East and North Africa.
- First-authored the resulting paper, published in *npj Clean Air* (2026).

**University of Moratuwa, Sri Lanka** **Sept 2023 – June 2024**
*Undergraduate Thesis Research Student* *Advisor: Dr. Chamira Edusooriya*

- Developed attention-based temporal convolutional models for motor imagery and P300 target/non-target EEG classification.
- Applied Grad-CAM to EEG scalograms to examine time-frequency regions contributing to classification decisions.
- Built an ear-EEG acquisition hardware prototype, including a fabricated printed circuit board and a 3D-printed earpiece.

**University of Sydney, Australia** **Jan 2023 – Sept 2023**
*Research Assistant - Internship* *Advisor: Dr. Anusha Vithana*

- Implemented ML-based structure prediction pipelines and compared intermediate model outputs against physics-based simulations to evaluate learned representations and real-world dynamical behavior.

## SELECTED HONORS AND AWARDS

**Bronze Medal**, International Mathematical Olympiad (IMO) 2017
**Honourable Mention**, International Mathematical Olympiad (IMO) 2014, 2015, 2016
**Bronze Medal**, Asian Pacific Mathematics Olympiad (APMO) 2018
**Honourable Mention**, International Mathematics Competition for University Students 2022
**Honourable Mention**, Asian Physics Olympiad (APhO) 2019
**Gold Medal**, Sri Lankan Physics Olympiad 2019

## OTHER SELECTED PROJECTS

**Vision-based Object Recognition and Task Solving Robot** **Feb 2022 – Aug 2022**

- Built a vision-guided robot navigation pipeline in Webots using camera inputs and OpenCV/C++, connecting visual perception outputs to autonomous task execution.

**AHEAD Intelligent Camera Project** **Dec 2022 – Mar 2023**

- Implemented real-time YOLO-based object detection and face recognition on an embedded Axis P1341 camera, optimizing vision inference for resource-constrained deployment.

**Protein Folding: Free Energy Landscape with ML** **2023**

- Implementing RoseTTAFold protein folding algorithm and taking the intermediate structures that a protein goes through while the algorithm is predicting the final structure.
- Using GROMACS Molecular Dynamics simulations to identify the folding pathways of a protein. Checking the similarities between the intermediate stages from two methods to identify how much of real world physics is learned by folding algorithms.

**Project Eyeguard** **Mar 2021 - Feb 2022**

- Developing a device to minimize the eye defects that can arise from long-term screen usage, by using cameras to track the eye and the blinking rates, and sending feedback to the users to take actions accordingly.

## SKILLS

- Python
- PyTorch
- TensorFlow
- MATLAB
- ROS

## RELEVANT COURSEWORK

**Computer Vision**: EN4553 Machine Vision (A+), EN4583 Advances in Machine Vision (A+), EN4573 Pattern Recognition and Machine Intelligence(A+)
**Signal Processing**: ENEE620 Random Processes in Communication and Control (A-), ENEE627 Information Theory (A-), ENEE630 Advanced Digital Signal Processing (A)
**Mathematics**: MA2023 Calculus (A+), MA2033 Linear Algebra (A), MA4013 Linear Models and Multivariate Statistics (A), MA2013 Differential Equations (A)

## LEADERSHIP/EXTRACURRICULARS

**President**: Mathematics Society, University of Moratuwa 2021
**Student Representative**: Department of Electronic and Telecommunication, University of Moratuwa 2022
**Committee Member**: Electronics Club, University of Moratuwa 2023
**Sergeant at Arms**: Gavel Club, University of Moratuwa 2022
**People Management Team**: AIESEC in Colombo South 2021
**Prefect**: Maliyadeva College 2018

## REFERENCES

**Prof. Dinesh Manocha**
Distinguished University Professor
Department of Computer Science
University of Maryland
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**Prof. Salman Khan**
Professor
Computer Vision Department
Mohamed bin Zayed University of Artificial Intelligence
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